

P09. Overtaking-enabled Eco-approach Control at Signalized Intersections for Connected and Automated Vehicles

中英双语博士精读笔记

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Theme / 主题: Eco-approach + relaxed FIFO

Method family / 方法族: Overtaking-enabled two-stage eco-approach / 允许超车的两阶段生态接近控制

PDF pages / 页数: 12 **Relevance / 相关度:** 84/100

Source / 来源: <https://arxiv.org/abs/2306.09736>

1 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

1.1 研究背景与痛点 / Background & Pain Points

这篇论文位于“Eco-approach + relaxed FIFO”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Eco-approach + relaxed FIFO. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Pain point / 痛点: Signalized intersection and overtaking/lane planning assumptions are outside the current fixed-route JSSP scope.

1.2 核心科学问题 / Core Scientific Question

在信号交叉口中，放松 FIFO 队列并允许超车，能否同时降低能耗和延误？

Can relaxing FIFO with overtaking-enabled lane optimization reduce both energy use and delay at signalized intersections?

1.3 科学贡献 / Scientific Contributions

- 方法贡献 (method contribution): Receding-horizon two-stage control: dynamic-programming lane optimization followed by PMP speed trajectory optimization.

- 安全/避碰贡献 (safety contribution): Handles dynamic disturbance from preceding vehicles and lane-choice constraints.
- 平滑/停止避免贡献 (smoothing contribution): Relaxes FIFO queuing at signalized intersections to reduce driving cost, energy consumption, and delay.
- 创新力度评判 (reviewer judgement): 较强工程洞察: 明确挑战 FIFO 保守性, 对 JSSP 重排序动机非常有用。

1.4 核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Overtaking-enabled two-stage eco-approach	允许超车的两阶段生态接近控制	该论文的核心方法族。

2 教材式学习路径与关键截图 / Textbook-Style Study Path & PDF Snapshots

2.1 学习路径 / Study Path

- 第一遍先读首页截图、摘要与引言, 确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading, 对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验, 判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

2.2 博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式: 车道 DP 与速度 PMP 的两阶段目标 / Two-stage lane-DP and speed-PMP objective, 并解释每个符号的物理意义?
- 能否画出输入-模型-算法-控制-验证的数据流图?
- 能否指出至少两个强假设, 并说明它们在真实交通中何时失效?
- 能否复述实验基准是否公平, 指标是否覆盖效率、安全、能耗和计算时间?
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验?

2.3 关键截图讲解 / PDF Snapshot Explanation

Screenshots are selected anchors from the local PDFs for study and parallel reading. They do not replace the original paper.

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Overtaking-enabled Eco-approach Control at Signalized Intersections for Connected and Automated Vehicles

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Abstract—Preceding vehicles typically dominate the movement of following vehicles in traffic systems, thereby significantly influencing the efficacy of eco-driving control that concentrates on vehicle speed optimization. To potentially mitigate the negative effect of preceding vehicles on eco-driving control at the signalized intersection, this paper proposes an overtaking-enabled eco-approach control (OEAC) strategy. It combines driving lane planning and speed optimization for connected and automated vehicles to relax the first-in-first-out queuing policy at the signalized intersection, minimizing the target vehicle's energy consumption and travel delay. The OEAC adopts a receding horizon two-stage control framework to derive optimal driving trajectories for adapting to dynamic traffic conditions. In the first stage, the driving lane optimization problem is formulated as a Markov decision process and solved using dynamic programming, which takes into account the uncertain disturbance from preceding vehicles. In the second stage, the vehicle's speed trajectory with the minimal driving cost is optimized rapidly using Pontryagin's minimum principle to obtain the closed-form analytical optimal solution. Extensive simulations are conducted to evaluate the effectiveness of the OEAC. The results show that the OEAC is excellent in driving cost reduction over constant speed and regular eco-approach and departure strategies in various traffic scenarios, with an average improvement of 20.91% and 5.62%, respectively.

Index Terms—Eco-driving; Connected and automated vehicles; Markov decision process; Pontryagin's minimum principle; Lane-changing; Speed optimization

I. INTRODUCTION

The growth of urbanization and the increasing vehicle ownership have led to an increase in the number of vehicles in urban traffic [1, 2], raising concerns about greenhouse gas emissions and traffic congestion [3]. Fortunately, by connecting vehicles and road infrastructure, the emerging

connected and automated vehicle (CAV) technologies provide possibilities to further improve vehicle energy efficiency and traffic throughput [4]. In this context, a variety of eco-friendly methods have been proposed from road or vehicle perspectives to improve vehicle and traffic efficiency. One such method is to optimize the traffic light signal phase and timing (SPaT) for prioritization of a single vehicle or a platoon approaching the intersection [5]. Another approach involves economically controlling the vehicle speed drive to pass through the signalized intersection, known as eco-approach and departure (EAD) control in some literature [6].

The EAD aims to facilitate smooth vehicle passage through signalized intersections, by avoiding stop-and-go behavior, and utilizing look-ahead information on traffic lights and preceding vehicles [6, 7]. A simple EAD is green light optimized speed advisory system, which uses deterministic traffic light SPaT, speed limits, and vehicle kinematics to calculate the speed range that ensures the vehicle can pass through an intersection without stopping [8]. Considering the uncertain SPaT of actuated traffic lights, Mahler *et al.* [9] introduced a prediction model to determine the probability of green signals at each time step, then the vehicle energy-optimal speed trajectory was derived that maximizes the likelihood of passing through an intersection. In real-world traffic, the vehicle speed may be impacted by the preceding vehicle or the queue waiting at the intersection. Therefore, Hao *et al.* [7], Bai *et al.* [10], and Ye *et al.* [11] designed the EAD strategies considering preceding vehicle movement information obtained using on-board radar, vehicle-to-vehicle communication, or prediction model. Dong *et al.* [12] and Sun *et al.* [13] proposed enhanced EAD to allow the vehicle to drive through the intersection without a stop by considering the discharging time of the vehicle queue, where the vehicle kinematics and traffic shockwave theory were utilized to estimate the vehicle queue movement. Other EAD strategies were proposed by Wang *et al.* [14], Han *et al.* [15], and Dong *et al.* [16]. These strategies are based on designing multiple intersection-based EAD controller that incorporates the spatial-temporal correlation among signalized corridors. Ma *et al.* [17], Li *et al.* [18], and Jiang *et al.* [19] proposed vehicle platoon-based EAD approaches that improve the energy efficiency of multiple vehicles passing through the signalized intersection.

Most EAD research has focused on optimizing vehicle speed to control its longitudinal movement. However, in some scenarios, the host vehicle (HV) may encounter slow-

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图 1: 首页与摘要 / Title and abstract page (p.1, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Supports the critique that FCFS/FIFO can be suboptimal and that speed smoothing should be coupled to sequencing.

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signal is lumped into the red for driving safety concerns.

The set of surrounding vehicles (SV) is given by $\mathcal{O}_s = \{N_{sv}, s_{s,j}, v_{s,j}, L_{s,j}\}$. Specifically, N_{sv} is the number of SVs, where we specify vehicle serial numbers as $j = 1, 2, \dots, N_{sv}$, in the order of proximity to the intersection and being from the innermost lane to outermost lane. $s_{s,j}$, $v_{s,j}$, and $L_{s,j}$ are the position, speed, and driving lane of j th SV, respectively. The SVs are classified into three types by their relative position with respect to the HV in the longitudinal direction: the vehicles in front of the HV are the preceding vehicles, the vehicles driving side by side with the HV are the side vehicles, and the vehicles behind the HV are the rear vehicles.

It is also assumed that the HV is equipped with a vehicle-to-infrastructure communication device (4G cellular network) and can access instantaneous traffic information. The HV can modify its speed in the communication zone and change the driving lane in the lane change coordinator zone. Note that the lane-changing rule is constructed to only permit the HV to drive in the current lane or change to the adjacent right or left lanes, as shown in Fig. 1. Therefore, the maximum permissible lane shift is only one, and the lane-changing index L_c has only three values, i.e., $L_c = [-1, 0, 1]$, corresponding to changing lanes to the left, driving in the current lane, and changing lanes to the right, respectively.

B. Vehicle Model

When changing lanes, we assume that the steering action is mild and the speed is moderate, so the vehicle complies with the dynamics and movement geometry constraints. Then, the bicycle model with an Ackerman steering design is used to explain vehicle lateral and longitudinal movements [28], as expressed in (1),

$$\begin{bmatrix} \dot{s}_x \\ \dot{s}_y \\ \dot{\gamma} \end{bmatrix} = \begin{bmatrix} \cos \gamma \\ \sin \gamma \\ \tan \beta_w / L \end{bmatrix} v \quad (1)$$

where v is the speed of HV, γ is the yaw angle, β_w is the front-wheel steering angle, L is the HV body length, and s_x and s_y are the HV's longitudinal and lateral positions, respectively.

We focus on the EAD control of an electric vehicle that is powered by a centralized electric motor. Ignoring the electric wire loss, the battery power is calculated by (2),

$$P_b = P_m \eta_b^{-\text{sign}(P_m)} + P_a \eta_b^{-1} \quad (2)$$

where η_b is the battery efficiency, P_a is the vehicle accessory power, and P_m is the motor power. The approximated closed-form expression of motor power [29] is used to reduce the computational burden, as expressed in (3),

$$P_m = \omega T_m v + \kappa T_m^2 \quad (3)$$

where ω and κ are the empirical coefficients, i.e., $\omega = i_g i_0 / r_w$ and $\kappa = R_m / \mu^2$. R_m is the motor armature resistance, μ is the motor speed constant, and T_m is the motor torque, as defined in (4) [30],

$$T_m = \frac{(mgf \cos \theta + mg \sin \theta + 0.5 C_D A \rho v^2 + m \delta v) r_w}{i_g i_0 \eta_t \text{sign}(T_m)} \quad (4)$$

where m is the vehicle mass, g is the gravity acceleration, f

is the rolling resistance coefficient, C_D is the aerodynamic drag coefficient, A is the frontal area, ρ is the air density, δ is the vehicle rotational inertia coefficient, r_w is the radius of the vehicle tire, i_g is the transmission ratio, i_0 is the final drive ratio, η_t is the driveline efficiency, θ is the road slope, and $\text{sign}(\cdot)$ is the signum function.

III. PROBLEM FORMULATION AND CONTROL FRAMEWORK

A. EAD Problem Analysis

EAD control is recognized as a promising approach to reducing vehicle energy consumption and travel delay in urban traffic at signalized intersections [4]. Regular EAD (READ) uses traffic light and preceding vehicle information, as shown in the blue solid line in Fig. 2, to calculate the vehicle speed to pass through the signalized intersection without stop-and-go operation. While approaching the preceding vehicle, the HV must adhere to the FIFO queuing policy, which may increase the travel time when encountering a slow-moving preceding vehicle. Additionally, due to the blocking effect of the preceding vehicle, the speed of READ may not be optimal.

In this context, to realize a more efficient vehicle control system, this paper proposes an overtaking-enabled EAD strategy, referred to as OEAC, for safely and efficiently passing through a signalized intersection by combining driving lane planning and speed optimization. As shown in the green solid line in Fig. 2, the HV with OEAC can overtake the slower vehicle, therefore relaxing the FIFO queuing policy to lower vehicle energy consumption and travel delay effectively.

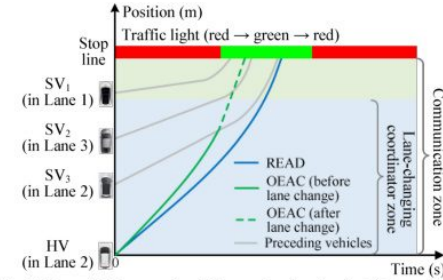


Fig. 2. Schematic diagram of a vehicle crossing the signalized intersection with READ and OEAC strategies. Here, the HV initially drives in Lane 2 before changing to Lane 1.

B. Optimization Control Problem

The objective of the proposed OEAC is to maneuver the vehicle efficiently passing through the signalized intersection with a minimal driving cost. The controller must determine the optimal driving lane and speed for the HV, considering relevant vehicle and traffic constraints, to reach the destination with the shortest amount of time and reduced energy consumption.

We define the state variable x as including vehicle speed and longitudinal position, i.e., $x = [v, s_x]^T$. To improve vehicle energy efficiency, we assume no mechanical brake

图 2: 建模或问题设定页 / Modeling or problem-setup page (p.3, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Supports the critique that FCFS/FIFO can be suboptimal and that speed smoothing should be coupled to sequencing.

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the constrained and unconstrained problems if the unconstrained optimal solution violates any of the constraints. The procedure is continued until no more constraints are triggered by the solution. However, the large number of repeated calculations raises the computational cost. In this context, the active conditions of state and control input constraints are implemented based on the findings of Dong *et.al* [40] and Mahbub *et.al* [41], to eliminate some constraints. For example, the control constraints only exist in the first phase of the optimal control sequence. Therefore, only state constraints rather than control input constraints may be activated later if the initial phase focuses on the unconstrained optimal solution.

The optimal control sequence depends on the order in which the vehicle state and control constraints are activated. For each case, the optimal control input, states, and entry time yield a set of two-point boundary value problem algebraic equations, which must be simultaneously solved while taking all constraints into consideration. Finally, a series of control input and states variable (i.e., u_s^* , v^* , and s_x^*) within the prediction horizon can be obtained by using PMP in each receding optimization.

In summary, in the z th receding optimization, the optimal control $\mathcal{T}_{opt,z}$, speed $\mathcal{V}_{opt,z}$, longitudinal position $\mathcal{S}_{xopt,z}$, lateral position $\mathcal{S}_{yopt,z}$ are derived based on the $\mathcal{L}_{opt,z}$. During the lane-changing process, $\mathcal{V}_{opt,z} = V_{h,z}$, $\mathcal{T}_{opt,z} = mgfr_w / i_g i_0 \eta_t$, and $\mathcal{S}_{xopt,z}$ and $\mathcal{S}_{yopt,z}$ are calculated by using vehicle kinematics and lane-changing trajectory models. In the lane-keeping process, the lateral position is not changed, therefore $\mathcal{S}_{yopt,z} = 0$. The derived optimal control input and state variables at the first step are then applied, that is, $\mathcal{T}_{mopt,z} = u_s^*(1)$, $\mathcal{V}_{opt,z} = v^*(1)$, and $\mathcal{S}_{xopt,z} = s_x^*(1)$.

V. SIMULATION AND RESULTS

A. Simulation Setup

The vehicle parameters are taken from the Chery Little-Ant, as listed in Table I. The coefficients that convert the energy consumption and travel time into the monetary counterpart are 0.12 USD/kWh and 24 USD/hour, respectively, according to the study in [42]. The traffic system is simulated using the real-world urban route as shown in Fig. 5, i.e., the intersection of Shuanglong Avenue and Jiyin Avenue, Nanjing, China, with the parameters listed in Table II. The IDM model is employed as the car-following model to imitate the driving behavior of vehicles, and the LC2013 model is used for describing lane-changing decisions [43]. The simulation is performed in MATLAB (version 9.13, 2022b) on a PC with Intel Core i9-12900k @ 3.20GHz CPU and 64GB RAM, the sampling time is 0.01s in simulation.

B. Benchmark Strategy

To evaluate the performance of the OEAC, the constant speed (CS) and READ strategies are used as benchmarks. In a free-driving situation, the CS strategy generally maintains a constant speed for the vehicle [12]. In addition, READ generates an energy-efficient speed profile using the PMP algorithm while only considering lane-keeping operation [40].

When a preceding vehicle appears in front of the HV, the car-following strategy is activated in both CS and READ to keep a safe distance. The car-following driving behavior is modeled by the IDM.

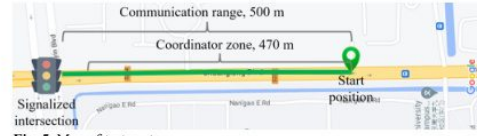


Fig. 5. Map of test route.

TABLE I
VEHICLE PARAMETERS.

Parameter	Value	Parameter	Value
Air density ρ	1.206kg/m ³	Tire radius r_w	0.280m
Air drag coefficient C_D	0.3	Transmission ratio i_g	2.80
Front area A	2.02m ²	Motor loss coefficient M_2	0.873
Final drive ratio i_0	3.789	Vehicle mass m	1005kg
Rolling resistance f	0.015	Auxiliary power P_a	300W
Rotational inertia δ	1.022	Gravity acceleration g	9.8m/s ²
Maximum acceleration a_{max}	2m/s ²	Motor maximum propulsion torque T_{mmax}	69Nm
Maximum deceleration a_{min}	-2m/s ²	Motor maximum generation torque T_{min}	-61Nm

TABLE II
ROADWAY NETWORK AND TRAFFIC PARAMETERS.

Parameter	Value	Parameter	Value
Location of stop line S_{tl}	500m	Width of lane D_w	3.5m
Communication range D_{cr}	500m	Time of red signal T_{gr}	51s
Coordinator zone D_{lr}	470m	Time of green signal T_{gr}	35s
Maximum speed V_{max}	70km/h	Standstill distance H_s	2m
Minimum speed V_{min}	20km/h	Safety time headway T_h	1.25s
Number of lanes N_l	3	Road slope θ	0

C. Simulation Results

1) *Performance Verification in Stochastic Traffic Scenarios:* To evaluate the driving cost reduction effectiveness of OEAC, a stochastic simulation is conducted. The HV is initially located in the middle lane, and 10000 individual simulation trials are conducted with constant S_{tl} , D_{cr} , D_{lr} , V_{min} , V_{max} , T_{gr} , and T_{re} . The parameters of the HV initial speed, traffic light initial states (i.e., I_{in} (0 or 1) and T_{is} ($1-T_{gr}$ or $1-T_{re}$)) traffic flow conditions (vehicle density (0 – 240 veh/km), and initial position and speed of SV) are all randomized in the simulation for a thorough investigation. Note that V_s is equal to the average traffic flow speed and is related to vehicle density.

The heat maps of the driving cost reduction of stochastic tests are depicted in Figs. 6 and 7. We point out that each grid of the heat map represents the average improvement of the different traffic scenarios with various speeds and positions of SVs. Table III compares the driving cost reduction and computational time of the OEAC in each step.

Figs. 6 and 7 display the distribution of driving cost reduction for the OEAC under various traffic conditions. The minimum and maximum driving cost reduction of the OEAC

图 3: 实验或结果页 / Experiments or results page (p.8, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Supports the critique that FCFS/FIFO can be suboptimal and that speed smoothing should be coupled to sequencing.

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- [31] S. Heydari, P. Fajri, R. Sabzehgar and A. Asrari, "Optimal brake allocation in electric vehicles for maximizing energy harvesting during braking," *IEEE Trans. Energy Convers.*, vol. 35, no. 4, pp. 1806-1814, Dec. 2020.
- [32] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Phys. Rev. E.*, vol. 62, no. 2, pp. 1805-1824, Aug. 2000.
- [33] Y. Liu, B. Zhou, X. Wang, L. Li, S. Cheng, Z. Chen, and L. Zhang, "Dynamic lane-changing trajectory planning for autonomous vehicles based on discrete global trajectory," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 8513-8527, July 2022.
- [34] R. E. Bellman, "A Markov decision process," *Journal of Mathematical Mechanics*, vol. 6, no. 5, pp. 679-684, 1957.
- [35] M. L. Puterman, "Markov decision processes: discrete stochastic dynamic programming," John Wiley & Sons, 2014.
- [36] Y. Xie et al., "Microsimulation of electric vehicle energy consumption and driving range," *Appl. Energy*, vol. 267, Jun. 2020, Art. no. 115081.
- [37] J. Han, J. Rios-Torres, A. Vahidi, and A. Sciarretta, "Impact of model simplification on optimal control of combustion engine and electric vehicles considering control input constraints," *Proc. IEEE Vehicle Power Propuls. Conf. (VPPC)*, Aug. 2018, pp. 1-6.
- [38] R. F. Hartl, S. P. Sethi, and V. G. Raymond, "A survey of the maximum principles for optimal control problems with state constraints," *SIAM Rev.*, vol. 37, no. 2, pp. 181-218, 1995.
- [39] B. Chalaki and A. A. Malikopoulos, "Optimal control of connected and automated vehicles at multiple adjacent intersections," *IEEE Trans. Control Syst. Technol.*, vol. 30, no. 3, pp. 972-984, May 2022.
- [40] H. Dong et al., "Event-driven energy-efficient driving control in urban traffic for connected electric vehicles," *IEEE Trans. Transport. Electric.*, vol. 9, no. 1, pp. 99-113, March 2023.
- [41] A. I. Mahbub and A. A. Malikopoulos, "Conditions to provable system-wide optimal coordination of connected and automated vehicles," *Automatica*, vol. 131, no. 109751, Sept. 2021.
- [42] T. Pan, R. Guo, W. H. Lam, R. Zhong, W. Wang, and B. He, "Integrated optimal control strategies for freeway traffic mixed with connected automated vehicles: a model-based reinforcement learning approach," *Transp. Res. Part C. Emerg. Technol.*, vol. 123, no. 102987, Feb. 2021.
- [43] Erdmann, "Lane-changing model in SUMO," *Proceedings of the SUMO User Conference Modeling Mobility with Open Data*, vol. 24, pp. 77-88, 2014.

图 4: 结论或局限页 / Conclusion or limitations page (p.12, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Supports the critique that FCFS/FIFO can be suboptimal and that speed smoothing should be coupled to sequencing.

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moving vehicles ahead. If the HV does not overtake the preceding vehicle, the HV speed optimization is restricted to the car-following mode, resulting in degraded efficacy of EAD control. Earlier vehicle overtaking research concentrated on lane-changing decision and trajectory optimization. Zhou *et al.* [20] and Li *et al.* [21] formulated optimal lane-changing strategies that consider HVs and other vehicles in their vicinity to improve driving comfort, safety, and lane-changing efficiency in complex driving environments. These studies, however, omit the vehicle energy-saving objective, which could lead to a reduction in vehicle energy efficiency when frequent lane changes are present. To address this issue, Gu *et al.* [22] have developed a joint control approach to simultaneously regulate vehicle longitudinal and lateral movement, allowing CAVs to travel in an energy-saving manner by adjusting the space headway and driving speed. Chen *et al.* [23] have derived time and energy-optimal lane-changing control policies that cooperate with neighboring CAVs.

The studies mentioned above focus on overtaking-enabled eco-cruising control on highways; however, the offered solutions may not apply to urban traffic due to the presence of traffic lights. Lane-changing in urban traffic is not only for achieving the goals of improving vehicle energy efficiency and reducing travel delay by considering the impact of preceding vehicles as in a highway scenario, but also to ensure vehicles pass through signalized intersections without stop-and-go operation during the green window. Currently, there are few investigations of overtaking-enabled EAD. Yang *et al.* [24] designed a less-disturbed EAD strategy that integrates offline energy-efficient speed planning and online speed tracking with overtaking ability. Nevertheless, the less-disturbed EAD does not achieve an energy-saving control globally since lane changing is not considered in vehicle speed planning. Guo *et al.* [25] developed a hybrid deep Q-learning and policy gradient-based EAD under a full penetration rate of CAVs environment, where both speed optimization and lane-changing order were considered. However, because a fully connected environment can only be realized in the distant future [26, 27] and the proposed approach does not take vehicle lane-changing trajectory into consideration, this approach is not practical in real-world traffic. Moreover, Hu *et al.* [28] proposed an enhanced EAD controller to cut through the traffic and catch green signals by overtaking slow-moving vehicles for real-time application in a partially connected environment. However, this study formulated a nonlinear optimal control problem based on a simplified vehicle model that ignored the vehicle powertrain model. The resulting solution could be different from reality.

This study proposes an overtaking-enabled eco-approach control (OEAC) strategy for CAVs, which is capable of relaxing the first-in-first-out (FIFO) queuing policy at a signalized intersection, while minimizing vehicle energy consumption and travel delay in a computationally efficient manner. The major contributions are threefold:

1) An overtaking-enabled EAD optimal control problem is formulated to cooperate with driving lane planning and speed optimization while considering the constraints posed by traffic lights and preceding vehicles. The unified monetary

counterpart objectives of energy consumption and travel time are used to balance vehicle energy and traffic efficiency.

2) A real time optimal driving trajectory is derived through a receding horizon two-stage control framework. In the first stage, the driving lane decision is formulated as a Markov decision process (MDP), and dynamic programming is adopted to solve this problem while considering the uncertain disturbance of preceding vehicles. In the second stage, the speed trajectory is optimized using Pontryagin's minimum principle (PMP) algorithm with minimal driving costs, leading to rapid optimization.

3) The simulation of stochastic traffic scenarios and typical case studies are investigated to verify the effectiveness of the proposed OEAC strategy. Furthermore, the sensitivity of OEAC's effectiveness to traffic flow conditions and traffic light initial states is inspected.

The remainder of this paper is organized as follows. Section II formulates the intersection crossing scenario and vehicle model. In Section III, the EAD optimal control problem and the OEAC framework are defined. Section IV provides the methodology of the OEAC in detail. The performance of the OEAC is comprehensively evaluated in Section V. Finally, Section VI concludes this paper.

II. INTERSECTION CROSSING SCENARIO AND VEHICLE MODEL

A. Intersection Crossing Scenario

We consider a general three-lane urban route with a fixed-timing traffic light, as shown in Fig. 1.

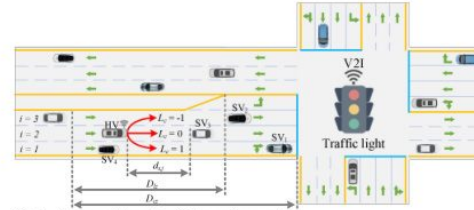


Fig. 1. A general signalized intersection route.

We define the route as a set $O_r = \{S_{tl}, D_{cz}, D_{lz}, N_l, D_w, V_{max}, V_{min}\}$. S_{tl} is the location of the stop line at the intersection, D_{cz} is the communication zone, D_{lz} is the lane-changing coordinator zone, which is related to D_{cz} and the length of no lane-changing zone near the intersection stop line. N_l is the total number of lanes in the same driving direction as the HV, where the lane numbers from the outside to the inside of the road are $i = 1, 2, \dots, N_l$. D_w is the width of the lane, and V_{max} and V_{min} are maximum and minimum speed limits, respectively.

The information on traffic lights is defined by a set $O_t = \{T_{ls}, I_{ln}, T_{gr}, T_{re}\}$, where T_{ls} is the initial transition time of the traffic light indication when the HV is approaching the communication zone, I_{ln} is the initial indication of the traffic light with $I_{ln} = 1$ and $I_{ln} = 0$ denoting the green and red signals, respectively, and T_{gr} and T_{re} are the time interval of green and red signals, respectively. Note that the yellow

图 5: 补充精读页 / Supplemental close-reading page (p.2, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Supports the critique that FCFS/FIFO can be suboptimal and that speed smoothing should be coupled to sequencing.

3 章节精读与逐段导读 / Section-by-Section Deep Reading & Walkthrough

p.1 I. INTRODUCTION

章节目标 / Learning goal: 建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 Eco-approach + relaxed FIFO：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决？

p.2 II. INTERSECTIONCROSSINGSCENARIO AND

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.3 III. PROBLEMFORMULATION ANDCONTROLFRAMEWORK

章节目标 / Learning goal: 把自然语言交通问题压缩为变量、状态、约束、目标函数或图结构。

Translate the traffic problem into variables, states, constraints, objectives, or graph relations.

关键论点 / Key argument: 本节是复现论文的核心入口，应与公式“车道 DP 与速度 PMP 的两阶段目标 / Two-stage lane-DP and speed-PMP objective”一起读。

技术焦点 / Technical focus: 逐项检查车辆状态、冲突关系、控制输入、时间/空间索引和安全阈值是否定义完整。

审稿追问 / Reviewer question: 这些建模假设在混合交通、通信延迟、感知误差和高密度车流下是否仍成立？

p.4 IV. OVERTAKING-ENABLEDECO-APPROACH

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 第一阶段用动态规划选择车道和超车策略，打破传统 FIFO；第二阶段用 PMP/最优控制生成速度曲线。滚动时域使策略能对前车扰动重新规划。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.8 V. SIMULATION AND RESULTS

章节目标 / Learning goal: 验证方法是否在公平基准下改善效率、安全、能耗或计算时间。

Check whether the claimed improvements are supported by fair baselines and meaningful metrics.

关键论点 / Key argument: 实验应与论文声称的贡献闭环：Extensive simulations report average driving-cost improvements over constant-speed and regular eco-approach baselines.

技术焦点 / Technical focus: 重点追踪指标：驾驶成本、能耗、延误、停车次数、超车次数和计算耗时。同时检查仿真平台、交通需求和随机性设置。

审稿追问 / Reviewer question: 实验是否只展示平均收益，还是覆盖了高密度、异常扰动和失败案例？

3.1 逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段 先把作者的痛点翻译成一句博士问题：在信号交叉口中，放松 FIFO 队列并允许超车，能否同时降低能耗和延误？读这一段时不要急着接受动机，要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段 把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括：车辆具备安全超车和变道能力。；信号相位和前车扰动可预测。；车道变化不会引入未建模的横向风险。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段 围绕核心公式“车道 DP 与速度 PMP 的两阶段目标 / Two-stage lane-DP and speed-PMP objective”复写符号表，确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段 把算法读成一条数据流：输入交通状态，执行 Receding-horizon two-stage control: dynamic-programming lane optimization followed by PMP speed trajectory optimization.，输出可执行通行/控制决策，并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段 不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：Extensive simulations report average driving-cost improvements over constant-speed and regular eco-approach baselines.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段 最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Supports the critique that FCFS/FIFO can be suboptimal and that speed smoothing should be coupled to sequencing.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

4 技术路线图与贡献量评估 / Technical Route & Contribution Matrix

Step	Stage / 阶段	Content / 内容
1	输入 / Input	车辆状态、路径/车道、交通需求、信号或协同信息。
2	建模 / Modeling	车道 DP 与速度 PMP 的两阶段目标 / Two-stage lane-DP and speed-PMP objective
3	核心求解 / Core solver	Receding-horizon two-stage control: dynamic-programming lane optimization followed by PMP speed trajectory optimization.
4	安全接口 / Safety interface	Handles dynamic disturbance from preceding vehicles and lane-choice constraints.
5	平滑与效率 / Smoothing and efficiency	Relaxes FIFO queuing at signalized intersections to reduce driving cost, energy consumption, and delay.
6	验证 / Validation	Extensive simulations report average driving-cost improvements over constant-speed and regular eco-approach baselines.

Dimension / 维度	Score	Comment / 评语
理论贡献 / Theoretical contribution	3/5	车道 DP 与速度 PMP 的两阶段目标 / Two-stage lane-DP and speed-PMP objective
算法贡献 / Algorithmic contribution	4/5	Receding-horizon two-stage control: dynamic-programming lane optimization followed by PMP speed trajectory optimization.

实验贡献 / Experimental contribution	3/5	Extensive simulations report average driving-cost improvements over constant-speed and regular eco-approach baselines.
工程贡献 / Engineering contribution	3/5	Handles dynamic disturbance from preceding vehicles and lane-choice constraints.
博士课题价值 / PhD relevance	4/5	Supports the critique that FCFS/FIFO can be suboptimal and that speed smoothing should be coupled to sequencing.

5 理论方法论深度解构 / Methodology & Theoretical Framework

5.1 问题数学建模 / Mathematical Formulation

车道 DP 与速度 PMP 的两阶段目标 / Two-stage lane-DP and speed-PMP objective

$$\min_{\ell_{0:H}, u(t)} J = \sum_{k=0}^H c_\ell(x_k, \ell_k) + \int_{t_0}^{t_f} c_v(v(t), u(t)) dt$$

Symbol / 符号	含义	Meaning	Role / 建模作用
ell_k	第 k 步车道选择	lane choice at step k	动态规划阶段决策
c_ell	车道/超车代价	lane/overtaking cost	平衡超车收益与扰动
u(t)	速度控制	speed-control input	连续速度优化变量
c_v	速度轨迹代价	speed-profile cost	包含能耗、时间和平滑性
H	滚动规划长度	receding-horizon length	在线计算窗口

5.2 核心算法架构 / Core Algorithm Layout

第一阶段用动态规划选择车道和超车策略，打破传统 FIFO；第二阶段用 PMP/最优控制生成速度曲线。滚动时域使策略能对前车扰动重新规划。

A receding-horizon controller first solves lane/overtaking choices with dynamic programming and then optimizes the speed profile with optimal-control tools.

5.3 收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

两阶段设计把组合车道决策和连续速度控制分开，实时性较好；但阶段间耦合不足时，车道选择可能导致后续速度优化不可行。

The decomposition improves tractability, but lane decisions can create downstream feasibility issues if speed constraints are not anticipated.

5.4 潜在假设与边界条件 / Assumptions & Boundary Conditions

- 车辆具备安全超车和变道能力。

- 信号相位和前车扰动可预测。
- 车道变化不会引入未建模的横向风险。

6 实验验证与结果评判 / Experimental Validation & Result Evaluation

Baselines & Environment / 基准与环境：常速接近、普通生态接近、FIFO 策略和不允许超车的控制器。

Validation / 验证：Extensive simulations report average driving-cost improvements over constant-speed and regular eco-approach baselines.

Metrics / 指标：驾驶成本、能耗、延误、停车次数、超车次数和计算耗时。

Ablation / 消融：需比较禁用超车、禁用 DP 车道优化和不同滚动时域长度。

Reviewer reading / 审稿式阅读提醒：阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

6.1 图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 24 figure references and 8 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
方法分类图	method-taxonomy diagram	把综述分类映射到你的 scheduler、smoother、safety filter 和 evaluation 四层框架。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

7 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

7.1 论文局限性 / Limitations & Weaknesses

- 信号场景与无信号 AIM 不完全一致。
- 超车可行性强依赖道路几何和混合交通行为。
- DP 状态空间随车道数和车辆数增长。

7.2 博士课题拓展点 / Research Extensions for PhD

- 把“允许超车”转化为 JSSP 中可交换相邻作业的局部搜索。
- 在无信号交叉口中研究 FIFO relaxation 的安全边界。
- 联合学习车道选择和排序，避免两阶段失配。

7.3 如何服务当前课题 / Use in This Project

Supports the critique that FCFS/FIFO can be suboptimal and that speed smoothing should be coupled to sequencing.

8 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数 / Parameter: FIFO 放松强度 / FIFO relaxation strength, range [0, 1] .
放松越强，潜在效率越高，但规则复杂度和可解释性风险增加。
Stronger FIFO relaxation can improve efficiency but raises rule complexity.

8.1 HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar：题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator，右侧为五大精读模块。
- 公式卡片使用 <pre> 或 LaTeX block，紧跟符号表。
- 审稿人批注用 reviewer-note block，博士拓展用 research-idea list。
- 交互参数用 input[type=range] + canvas 曲线，打印 PDF 时隐藏交互控件但保留解释。

9 来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

Page	Extracted heading
p.1	I. INTRODUCTION
p.3	III. PROBLEMFORMULATION ANDCONTROLFRAMEWORK
p.8	V. SIMULATION ANDRESULTS

p.1 I. INTRODUCTION 定位问题背景、研究缺口和为什么现有保守/非协同方法不足。/ Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.2 II. INTERSECTIONCROSSINGSCENARIO AND 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.3 III. PROBLEM FORMULATION AND CONTROL FRAMEWORK 把交通交互转写为状态、约束、目标或图结构，是精读时最应复现的部分。/ Defines states, constraints, objectives, or graph structures; this is the section to reproduce carefully.

p.4 IV. OVERTAKING-ENABLED ECO-APPROACH 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.8 V. SIMULATION AND RESULTS 检验方法是否真的改善效率、安全或能耗；要关注基准是否公平。/ Tests efficiency, safety, or energy claims; inspect whether baselines are fair.